# AIR POLLUTION AND CONTROL

## Subject Code : 23A01504b

## UNIT - I : AIR POLLUTION

**Syllabus:** Definition - Sources & Classification of Air Pollutants - Effects on Humans, Plants and Materials - Global Effects - Air Quality and NAAQS - National Clean Air Programme - Sampling of Pollutants in Ambient Air - Stack Sampling.

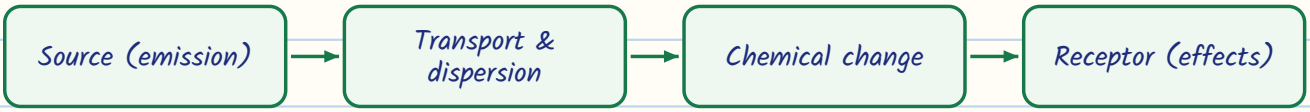
## 1. Air Pollution - Definition

Air is a mixture of gases: $N_2$ (78%), $O_2$ (21%), Ar (0.93%), $CO_2$ (0.03-0.04%) and traces of other gases plus water vapour. When the natural composition of air is disturbed, we call it pollution.

**DEFINITION: Air pollution is the presence in the outdoor atmosphere of one or more contaminants (dust, fumes, gas, mist, odour, smoke, vapour) in such quantity and of such duration that they are or may tend to be injurious to human, plant or animal life, or to property, or which unreasonably interfere with the comfortable enjoyment of life.**

**Air Act, 1981 (India):** 'Air pollutant' means any solid, liquid or gaseous substance (including noise) present in the atmosphere in such concentration as may be or tend to be injurious to human beings, other living creatures, plants, property or environment.

- Air pollution = Source + Atmosphere (transport, dilution, chemical change) + Receptor
- Pollutants are measured in $ug/m^3$, $mg/m^3$ (mass/volume) or ppm, ppb (volume/volume)
- Conversion: $ug/m^3$ = (ppm x mol. wt. x 1000) / 24.45 at 25 deg C and 1 atm




## 2. Sources of Air Pollutants

### (a) Natural sources

- Volcanic eruptions - ash, $SO_2$, $H_2S$, $CO_2$
- Forest fires - smoke, CO, $CO_2$, particulates
- Dust storms and wind-blown soil; sea spray (salt particles)
- Pollen, spores, bacteria; decay of organic matter (methane, $H_2S$)
- Lightning - forms NO from $N_2$ and $O_2$; cosmic dust, meteors

### (b) Anthropogenic (man-made) sources

- **Combustion of fuels** - thermal power plants, industries, domestic cooking and heating
- **Transportation** - cars, trucks, buses, ships, aircraft, railways (CO, $NO_x$, HC, Pb, PM)
- **Industrial processes** - cement, steel, fertiliser, refineries, chemical, paper, smelters
- **Agriculture** - pesticide spray, burning of crop residue, $NH_3$ from fertilisers, $CH_4$ from paddy and cattle
- **Solid waste** - open burning, incinerators, landfill gas
- **Miscellaneous** - construction, mining, quarrying, wars and nuclear explosions

### (c) Classification of sources by geometry

| Type | Description | Examples |
|---|---|---|
| Point source | Single fixed, identifiable location | Chimney, stack of power plant |
| Line source | Emission along a line / road | Highway, railway track |
| Area source | Many small sources over a region | City, domestic burning, landfill |
| Mobile source | Moves from place to place | Vehicles, ships, aircraft |
| Stationary source | Fixed in position | Factories, boilers, furnaces |

## 3. Classification of Air Pollutants

### (i) Based on origin

- **Primary pollutants** - emitted directly from the source. e.g. $SO_2$, NO, CO, soot, fly ash, Pb, HC
- **Secondary pollutants** - formed in the atmosphere by reaction of primary pollutants. e.g. ozone ($O_3$), PAN (peroxyacetyl nitrate), sulphuric acid mist, nitric acid, aldehydes

**Remember: Smog (photochemical) = $NO_x$ + HC + sunlight -> $O_3$ + PAN + aldehydes (all secondary)**

### (ii) Based on physical state

- **Gaseous pollutants** - $SO_2$, $NO_x$, CO, $CO_2$, $H_2S$, $NH_3$, HF, $O_3$, VOCs
- **Particulate pollutants** - solids or liquid droplets: dust, smoke, fumes, mist, fog, fly ash, aerosols

## (iii) Particulate matter - terms

| Term | Meaning |
|---|---|
| Dust | Solid particles by crushing/grinding, size > 1 um (settle by gravity) |
| Smoke | Fine solid/liquid particles from incomplete combustion, < 1 um |
| Fume | Solid particles by condensation of vapours (metal fumes), 0.03-0.3 um |
| Mist / Fog | Liquid droplets by condensation or atomisation; fog is visible mist |
| Fly ash | Non-combustible residue carried by flue gas of coal furnaces |
| Aerosol | Any suspension of fine solid/liquid particles in gas |
| PM10 / PM2.5 | Particles with aerodynamic diameter < 10 um / < 2.5 um |

## (iv) Based on chemical nature

- Inorganic: oxides of S, N, C; halogens; metals (Pb, Hg, Cd)
- Organic: hydrocarbons, aldehydes, ketones, PAHs, benzene, VOCs
- Radioactive: radon, iodine-131, strontium-90

## (v) Criteria pollutants

Pollutants for which standards are fixed on the basis of health criteria: **$SO_2$, $NO_2$, CO, $O_3$, Pb, PM10 and PM2.5** (US-EPA list of six; India's NAAQS covers 12).

# 4. Major Pollutants - Sources and Effects

| Pollutant | Main sources | Main effects |
|---|---|---|
| CO | Vehicle exhaust, incomplete combustion | Binds Hb (carboxyhaemoglobin), headache, death |
| $SO_2$ | Coal / oil burning, smelters | Bronchitis, acid rain, leaf damage |
| $NO_x$ | Vehicles, power plants (high temp.) | Lung irritation, smog, acid rain |
| Hydrocarbons | Vehicles, solvents, refineries | Smog, cancer (benzene, PAH) |
| Lead | Leaded petrol, battery, paint | Brain damage in children, anaemia |
| $O_3$ | Secondary (photochemical) | Eye/lung irritation, crop damage |
| Particulates | Industry, dust, vehicles, burning | Asthma, silicosis, visibility loss |

# 5. Effects of Air Pollution

## 5.1 Effects on human health

Pollutants enter the body mainly by **inhalation**. The nose filters particles > 10 um; particles of 2.5-10 um reach the bronchi; PM2.5 and smaller reach the alveoli and may even enter the blood.

- **Carbon monoxide:** affinity for haemoglobin is about 200-250 times that of oxygen. COHb reduces $O_2$ carrying capacity -> headache, dizziness, coma, death at ~ 1000 ppm.
- **$SO_2$:** irritates the upper respiratory tract; chronic bronchitis, asthma, emphysema. It combines with PM and the effect is synergistic.
- **$NO_2$:** deep lung irritant; pulmonary oedema; reduces resistance to infection; forms nitrosamines.
- **Particulates:** silicosis (silica), asbestosis, black lung (coal), byssinosis (cotton), lung cancer, heart disease.

- **Lead:** damages nervous system, kidney; lowers IQ in children. **Benzene:** leukaemia.
- **Ozone / PAN:** eye irritation, cough, chest pain, reduced lung function.
- **Fluorides:** fluorosis of bones and teeth. **Mercury/Cadmium:** Minamata and Itai-itai disease.

**Famous episodes: Meuse Valley (1930), Donora (1948), London smog (1952, ~4000 deaths), Los Angeles smog (1940s), Bhopal gas tragedy (Dec 1984, methyl isocyanate - MIC).**

## 5.2 Effects on plants

- **$SO_2$** - chlorosis (yellowing) and necrosis between veins; bleached spots. Sensitive: alfalfa, barley, cotton
- **Ozone** - flecking and stippling on upper leaf surface; tobacco, spinach, tomato are sensitive
- **PAN** - glazing / silvering on lower leaf surface; young leaves most affected
- **Fluorides** - tip burn, leaf margin necrosis; very toxic to plants even at ppb level (gladiolus, apricot)
- **Ethylene** - leaf fall (abscission), withered flowers; $NO_2$ - growth suppression
- **Particulates** - dust layer blocks stomata and sunlight, reducing photosynthesis (cement dust)
- Acid rain damages leaves and leaches nutrients from soil; forest die-back

## 5.3 Effects on materials

- **Metals:** $SO_2$ + moisture -> $H_2SO_4$; corrosion of iron, zinc, copper, aluminium
- **Building stones:** limestone, marble: $CaCO_3 + H_2SO_4 \rightarrow CaSO_4 + H_2O + CO_2$ (the Taj Mahal 'stone cancer')

- **Paints:** darkening and blistering by $H_2S$ (lead paint turns black as PbS)
- **Textiles, paper, leather:** weakened and discoloured by acid gases
- **Rubber:** cracks due to ozone (tyres, hoses); dyes fade due to $NO_x$
- Soiling of buildings by soot and dust: heavy cleaning cost

### 5.4 Effects on animals and atmosphere

- Fluorosis in cattle, arsenic and lead poisoning through grazing on contaminated grass
- Reduction in visibility (haze), less sunlight, change in local climate (urban heat island)

## 6. Global Effects of Air Pollution

### 6.1 Greenhouse effect and global warming

Short-wave solar radiation reaches earth; the earth re-radiates long-wave infrared, which is absorbed by **greenhouse gases (GHGs)** - $CO_2$, $CH_4$, $N_2O$, CFCs, $O_3$, water vapour - and radiated back. Natural greenhouse effect keeps earth at +15 deg C instead of -18 deg C. Enhanced GHG levels cause **global warming**.

| GHG | Main source | Relative GWP (100 yr) |
|---|---|---|
| $CO_2$ | Fossil fuel burning, deforestation | 1 |
| $CH_4$ | Paddy, cattle, landfills, coal mines | about 28 |
| $N_2O$ | Fertilisers, biomass burning | about 265 |
| CFC-12 / HFCs | Refrigerants, aerosols | thousands (about 10,000) |
| $SF_6$ | Electrical industry | about 23,500 |

**Consequences:** sea-level rise, melting of glaciers, change in rainfall, droughts, floods, spread of diseases, loss of biodiversity, crop yield changes.

**Control:** Kyoto Protocol (1997), Paris Agreement (2015) - hold warming well below 2 deg C, preferably 1.5 deg C; renewable energy, afforestation, energy efficiency.

## 6.2 Ozone layer depletion

Stratospheric ozone (15-35 km) absorbs UV-B (280-315 nm). **Ozone depleting substances (ODS):** CFCs, halons, $CCl_4$, methyl chloroform, HCFCs.

UV splits CFC in the stratosphere releasing Cl atoms which act as a catalyst:

- $CF_2Cl_2 + UV \rightarrow CF_2Cl + Cl$
- $Cl + O_3 \rightarrow ClO + O_2$
- $ClO + O \rightarrow Cl + O_2$ (Cl is regenerated; one Cl destroys about 100,000 $O_3$ molecules)

**Ozone hole** over Antarctica appears in September-October due to polar stratospheric clouds. **Effects:** skin cancer, cataract, weak immunity, damage to phytoplankton and crops. **Dobson Unit (DU)** measures ozone column; the 'hole' is < 220 DU.

**Montreal Protocol (1987) phases out ODS; Kigali Amendment (2016) phases down HFCs.**

## 6.3 Acid rain

Rain with **pH < 5.6** (normal rain is slightly acidic due to dissolved $CO_2$). $SO_2$ and $NO_x$ are oxidised and dissolve in cloud water:

- $SO_2 + OH \rightarrow HOSO_2$; $SO_2 + 1/2\ O_2 + H_2O \rightarrow H_2SO_4$
- $NO_2 + OH \rightarrow HNO_3$
- **Effects:** acidification of lakes (fish die), forest decline, soil nutrient loss (Ca, Mg), corrosion of buildings and statues, human health

- **Control:** flue gas desulphurisation, low-sulphur fuels, catalytic converters, liming of lakes

### 6.4 Photochemical smog and other global issues

- Photochemical smog (Los Angeles type): $NO_2$ + sunlight -> NO + O; O + $O_2$ -> $O_3$; HC + $O_3$ -> PAN, aldehydes
- Classical (London) smog: smoke + fog + $SO_2$; cold, humid, reducing conditions
- Atmospheric brown cloud (ABC), black carbon on Himalayan snow -> faster melting
- Transboundary pollution, haze and loss of visibility

| Feature | London smog | Photochemical (LA) smog |
|---|---|---|
| Season / time | Winter, early morning | Summer, noon |
| Temperature | Low (-1 to 4 deg C) | High (24-32 deg C) |
| Main pollutants | Smoke, $SO_2$, fog | $NO_x$, HC, $O_3$, PAN |
| Nature | Reducing | Oxidising |
| Effect | Bronchitis, deaths | Eye irritation, crop damage |

## 7. Air Quality and NAAQS

**Ambient air quality** is judged against standards. **National Ambient Air Quality Standards (NAAQS)** are limits of pollutant concentration in outdoor air, set by the **CPCB** under Air Act 1981 (first in 1994, revised 2009, notified Nov 2009). Standards give two categories of area: (i) industrial, residential, rural and other areas and (ii) ecologically sensitive areas (notified by Central Government).

| Pollutant | Time-weighted avg. | Industrial/Residential | Eco-sensitive |
|---|---|---|---|
| $SO_2$ | Annual / 24 hr | 50 / 80 ug/$m^3$ | 20 / 80 |
| $NO_2$ | Annual / 24 hr | 40 / 80 ug/$m^3$ | 30 / 80 |

| Pollutant | Time-weighted avg. | Industrial/Residential | Eco-sensitive |
|---|---|---|---|
| PM10 | Annual / 24 hr | 60 / 100 $ug/m^3$ | 60 / 100 |
| PM2.5 | Annual / 24 hr | 40 / 60 $ug/m^3$ | 40 / 60 |
| Ozone | 8 hr / 1 hr | 100 / 180 $ug/m^3$ | 100 / 180 |
| Lead | Annual / 24 hr | 0.5 / 1.0 $ug/m^3$ | 0.5 / 1.0 |
| CO | 8 hr / 1 hr | 2 / 4 $mg/m^3$ | 2 / 4 |
| $NH_3$ | Annual / 24 hr | 100 / 400 $ug/m^3$ | 100 / 400 |

Others: Benzene 5 $ug/m^3$ (annual), Benzo(a)pyrene 1 $ng/m^3$, Arsenic 6 $ng/m^3$, Nickel 20 $ng/m^3$ (annual) - total 12 parameters.

### Air Quality Index (AQI) - India, launched Oct 2014

- Converts concentration of 8 pollutants (PM10, PM2.5, $NO_2$, $SO_2$, CO, $O_3$, $NH_3$, Pb) into one number 0-500
- Good 0-50 | Satisfactory 51-100 | Moderate 101-200 | Poor 201-300 | Very Poor 301-400 | Severe 401-500
- Sub-index calculated for each pollutant; **AQI = maximum sub-index** (at least 3 pollutants, one of them PM10 or PM2.5)

**Air Quality Management:** monitoring -> emission inventory -> standards -> modelling -> control strategy -> enforcement. WHO global guidelines (2021): PM2.5 annual 5 $ug/m^3$, 24 hr 15 $ug/m^3$.

## 8. National Clean Air Programme (NCAP)

Launched by the **MoEFCC in January 2019** - the first national, time-bound, city-specific plan to curb air pollution in India.

- **Goal:** 20-30% reduction in PM10 and PM2.5 concentration by 2024 (base year 2017); later revised to **40% reduction by 2025-26**
- **Target cities:** 131 non-attainment cities (cities that did not meet NAAQS for 5 years, 2011-2015) - includes cities in 24 states/UTs

- Each city prepares a **City Action Plan** - road dust, vehicular, industrial, biomass, construction, waste burning
- Plan is implemented through **city / state / central level committees**; PRANA portal for tracking
- Funding: Finance Commission grants for million-plus cities; National Clean Air Fund concept
- Strengthening of **monitoring network** (CAAQMS, manual stations) and source apportionment studies
- Public awareness, capacity building, technology assessment, emission inventory, health impact studies
- Complementary steps: BS-VI fuels (April 2020), GRAP in Delhi-NCR, Ujjwala LPG scheme, Pollution Under Control (PUC) certificates, EV promotion (FAME), ban on stubble burning, SAFAR forecasting

## 9. Sampling of Pollutants in Ambient Air

**Objectives:** to find the quality of ambient air, to compare with NAAQS, to identify trends and sources, to assess health effects and the effectiveness of control and to plan siting of industries.

### Principles of sampling

- Samples must be **representative** - collected at 1.5-10 m above ground (breathing zone), away from local obstructions (2 m from walls, trees)
- Number and location of stations depend on population, topography, wind rose, industrial areas, pollutant type
- Sampling duration: 24 hours (4-hourly or 8-hourly for gases), minimum 104 measurements per year (twice a week)
- **Grab sampling** gives instantaneous values; **continuous sampling** gives averages and trends

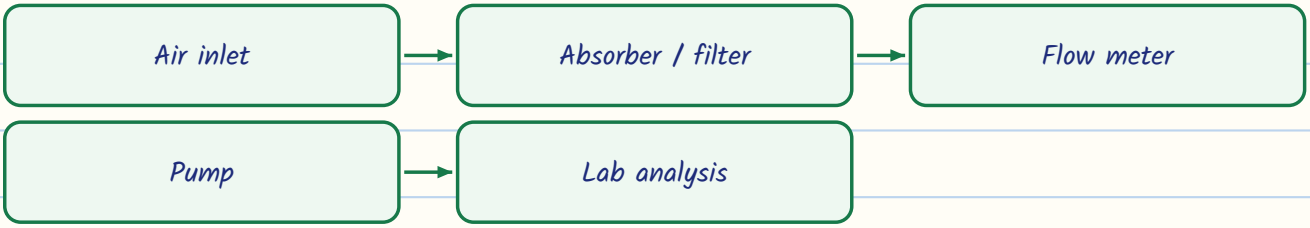
## (a) Particulate sampling - High Volume Sampler (HVS)

- Air drawn by a blower at **1.1-1.7 $m^3$/min** through a pre-weighed **glass-fibre filter** (20 x 25 cm) for 24 hr
- Mass concentration = $(W_2 - W_1) / V$ where V = flow rate x time (corrected to standard conditions)
- Collects SPM up to 100 um. For PM10 a **cyclone / impactor head** is fitted (Respirable Dust Sampler, RDS, flow 1.0-1.2 $m^3$/min)
- Filter weighed after conditioning in desiccator; also analysed for metals, sulphate, nitrate, BaP
- Others: Beta attenuation monitor (BAM), TEOM (oscillating microbalance), light scattering - **continuous**

## (b) Gaseous sampling

Gas is collected by bubbling through an absorbing solution in an **impinger / bubbler**, by adsorption on a solid (charcoal, silica gel), by condensation (cold trap) or **grab** in bags/bottles.

| Pollutant | Method | Absorbing reagent / principle |
|---|---|---|
| $SO_2$ | West-Gaeke (pararosaniline) | Sodium tetrachloromercurate; red-violet colour at 560 nm |
| $NO_2$ | Jacob-Hochheiser (modified) | Sodium hydroxide + sodium arsenite; azo dye, 540 nm |
| CO | Non-dispersive IR (NDIR) | IR absorption at 4.6 um |
| $O_3$ | Chemiluminescence / UV photometry | Reaction with ethylene or UV absorption 254 nm |
| $NH_3$ | Indophenol blue method | Dilute sulphuric acid, 630 nm |
| Pb | AAS / ICP | Extraction of filter paper in nitric acid |
| Hydrocarbons | Flame ionisation detector, GC | Carbon ions in hydrogen flame |




**Automatic continuous analysers (CAAQMS):** UV fluorescence ($SO_2$), chemiluminescence ($NO_x$), NDIR (CO), UV photometer ($O_3$) - data logged and sent online to CPCB.

## 10. Stack Sampling

Sampling of **flue gas inside a chimney/stack/duct** to measure the emission of particulate and gaseous pollutants from a source and to check compliance with emission standards (e.g. 150 mg/$Nm^3$ for particulates, as per CPCB).

### Why / when

- To check emission standards; to determine efficiency of control equipment; for design of control device; to find mass emission rate (kg/h)

### Sampling location (ports)

- In a straight vertical duct, **8 stack diameters downstream** and **2 diameters upstream** of any bend, fan, damper or obstruction
- Sampling ports (10 cm dia) at 90 deg apart; a sampling platform with safe access is needed
- **Traverse points:** the cross section is divided into equal areas (centre of each); small stacks need fewer points (minimum 8 for circular ducts up to 3 m dia)
- Velocity measured at every traverse point because velocity varies across the section

## Isokinetic sampling

**Isokinetic: the velocity of gas entering the sampling nozzle = velocity of gas in the stack at that point ($V_n = V_s$)**

- If $V_n > V_s$ (over-sampling): extra gas is drawn in, but heavy particles cannot turn and are missed -> **coarse particle conc. reads LOW**
- If $V_n < V_s$ (under-sampling): gas streams deflect around the nozzle but heavy particles go straight in by inertia -> **coarse particle conc. reads HIGH**
- Fine particles (< 5 um) follow gas streamlines, so error is small; isokinetic sampling gives a representative dust load

## Stack sampling train (EPA method 5 type)




1. **Nozzle & probe:** stainless steel/ glass-lined, nozzle sharp-edged, faced into the gas flow
2. **Pitot tube (S-type) + manometer:** measures velocity pressure
3. **Thermocouple:** measures stack gas temperature
4. **Filter holder (heated box):** collects particulate (at about 120 deg C to avoid condensation)
5. **Impingers:** collect moisture and gaseous pollutants ($SO_2$, $NO_x$) in absorbing solutions; silica gel dries the gas
6. **Pump, dry gas meter and orifice meter:** control and record sample volume; flow adjusted to remain isokinetic

### Calculations

- Stack gas velocity: $V_s = C_p \times \text{sqrt}( 2 \times g \times dP / \text{rho} )$ [$C_p$ = pitot coefficient (about 0.84 for S-type, 1.0 for standard)]
- Flow rate: $Q = V_s \times A$ ($m^3/s$); convert to normal conditions (0 deg C, 1 atm - $Nm^3$)
- Concentration C = mass of particulate on filter / volume of gas sampled (corrected to $Nm^3$, dry, 12% $CO_2$ or 7% $O_2$ reference)
- Emission rate (kg/h) = C x Q

**Gaseous stack sampling** is done by extractive probes with analysers (Orsat apparatus for $CO_2$, $O_2$, CO; flue gas analyser using electrochemical cells; Draeger tubes) or by in-situ / CEMS continuous emission monitoring.

**Ambient sampling -> measures air quality (low conc., ug/$m^3$). Stack sampling -> measures emissions (high conc., mg/$Nm^3$), needs isokinetic conditions.**

## II. Quick Revision / Important Questions

1. Define air pollution. Classify air pollutants with examples.
2. Explain the effects of air pollution on humans, plants and materials.
3. Write briefly on greenhouse effect, ozone depletion and acid rain.
4. What is NAAQS? List the standards for $SO_2$, $NO_2$, PM10 and PM2.5.
5. Describe the National Clean Air Programme (NCAP) and its objectives.
6. With a neat sketch describe the high-volume sampler.
7. What is isokinetic sampling? Explain a stack sampling train.